Exposure to diethylstilbestrol during sensitive life stages: a legacy of heritable health effects.

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Diethylstilbestrol (DES) is a potent estrogen mimic that was predominantly used from the 1940s to the 1970s by pregnant women in hopes of preventing miscarriage. Decades later, DES is known to enhance breast cancer risk in exposed women and cause a variety of birth-related adverse outcomes in their daughters such as spontaneous abortion, second trimester pregnancy loss, preterm delivery, stillbirth, and neonatal death. Additionally, children exposed to DES in utero suffer from sub/infertility and cancer of reproductive tissues. DES is a pinnacle compound that demonstrates the fetal basis of adult disease. The mechanisms of cancer and endocrine disruption induced by DES are not fully understood. Future studies should focus on common target tissue pathways affected and the health of the DES grandchildren.

[Diethylstilbestrol exposure in utero. Polemics about metroplasty. The pros].

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The diethylstilbestrol (DES) is a synthetic estrogen which was prescribed from 1941 onwards for the prevention of miscarriage. As well as a possible risk of cancer, another side effect of this treatment was the possible abnormality of the genitalia in the female issue of the prescribed user. Apart from possibly having a hypoplasic uterus, the patient is also prone, in the case where she has an undersized uterus, to having a much narrower than normal cavity. Consequently, there is a tendency for an excess of muscle tissue on the uterus walls. This can be observed on a RMN. The most significant characteristics of this abnormality are: constriction rings around the proximal uterine segment, a T shaped uterus, uterus with an arched base. The idea of the plastic enlargement operation (metroplasty) is to widen the cavity by making careful incisions of the excess muscle tissue located on the uterus wall. The objective of this is to obtain a triangular shaped cavity taking care though to weaken the walls themselves. 61 patients were treated. We observed 37 pregnancies after 16 months with 30 ongoing pregnancies. Generally, the anatomic results are excellent but it is difficult to measure the functional results of the success rate in future pregnancies. The reason for this is the enlarging of the cavity alone does not guarantee successful fertility. There are other problems to take into account e.g. implantation, miscarriage and premature labor. There are risks with this operation, such as placenta percreta, a possible rupture of the uterus, though this can happen at any time with DES patients. This operation can only be recommended once a thorough examination of the patient has been made.

Diethylstilbestrol exposure.

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Diethylstilbestrol is a synthetic nonsteroidal estrogen that was used to prevent miscarriage and other pregnancy complications between 1938 and 1971 in the United States. In 1971, the U.S. Food and Drug Administration issued a warning about the use of diethylstilbestrol during pregnancy after a relationship between exposure to this synthetic estrogen and the development of clear cell adenocarcinoma of the vagina and cervix was found in young women whose mothers had taken diethylstilbestrol while they were pregnant. Although diethylstilbestrol has not been given to pregnant women in the United States for more than 30 years, its effects continue to be seen. Women who took diethylstilbestrol during pregnancy have a slightly higher risk of breast cancer than the general population and therefore should be encouraged to have regular mammography. Women who were exposed to diethylstilbestrol in utero may have structural reproductive tract anomalies, an increased infertility rate, and poor pregnancy outcomes. However, the majority of these women have been able to deliver successfully. Recommendations for gynecologic examinations include vaginal and cervical digital palpation, which may provide the only evidence of clear cell adenocarcinoma. Initial colposcopic examination should be considered; if the findings are abnormal, colposcopy should be repeated annually. If the initial colposcopic examination is normal, annual cervical and vaginal cytology is recommended. Because of the higher risk of spontaneous abortion, ectopic pregnancy, and preterm delivery, obstetric consultation may be required for pregnant women who had in utero diethylstilbestrol exposure. The male offspring of women who took diethylstilbestrol during pregnancy have an increased incidence of genital abnormalities and a possibly increased risk of prostate and testicular cancer. Routine prostate cancer screening and testicular self-examination should be encouraged.

Psychosexual characteristics of men and women exposed prenatally to diethylstilbestrol.

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Between 1939 and the 1960s, the synthetic estrogen diethylstilbestrol (DES) was given to millions of pregnant women to prevent pregnancy complications and losses. The adverse effects of prenatal exposure on the genitourinary tract in men and the reproductive tract in women are well established, but the possible effects on psychosexual characteristics remain largely unknown. We evaluated DES exposure in relation to psychosexual outcomes in a cohort of 2,684 men and 5,686 women with documented exposure status. In men, DES was unrelated to the likelihood of ever having been married, age at first intercourse, number of sexual partners, and having had a same-sex sexual partner in adulthood. DES-exposed women, compared with the unexposed, were slightly more likely to have ever married (odds ratio [OR] = 1.1; confidence interval [CI] = 1.0-1.4) and less likely to report having had a same-sex sexual partner (OR = 0.7; CI = 0.5-1.0). The DES-exposed women were less likely to have had first sexual intercourse before age 17 (OR = 0.7; CI = 0.6-0.9) or to have had more than one sexual partner (OR = 0.8; CI = 0.7-0.9). There was an excess of left-handedness in DES-exposed men (OR = 1.4; CI = 1.1-1.7) but not in DES-exposed women. DES exposure was unrelated to self-reported history of mental illness in women. Overall, our findings provide little support for the hypothesis that prenatal exposure to DES influences the psychosexual characteristics of adult men and women.

Diethylstilbestrol: Potential health risks for women exposed in utero and their offspring.

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Diethylstilbestrol (DES) is a synthetic estrogen given to pregnant women to prevent miscarriages and preterm labor; the drug was used between 1941 and 1971 in the United States and into the 1980s in other countries. DES exposure is associated with significant long-term health effects, including increased risk for breast cancer, cervical and vaginal clear cell adenocarcinoma, reproductive tract abnormalities,

infertility, poor pregnancy outcomes, and early menopause. This article reviews the potential health risks associated with DES exposure, how to assess which patients are at risk, and management recommendations for patients exposed to DES.

Current perspective of diethylstilbestrol (DES) exposure in mothers and offspring.

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Diethylstilbestrol (DES) was an orally active estrogen prescribed to the pregnant women to prevent miscarriages. DES is known as a 'biological time bomb' and long-term effects of DES have been recorded in the mothers exposed to DES and their offspring (DES-daughters and DES-sons). Adverse pregnancy outcomes, infertility, cancer, and early menopause have been discovered in women exposed to DES, and some events occur in their offspring and subsequent generations. An increased risk of breast cancer is not limited to the DES-exposed daughters. There is an urgent need to find ways to stop the inheritance cycle of DES and prevent adverse effects of DES in the future generations. The present article reviews the health implications of DES exposure and screening exams currently recommended to DES daughters and their offspring.

Clinical implications of DES.

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Once prescribed to pregnant women to prevent risk of spontaneous abortion, diethylstilbestrol (DES) is now associated with an increased risk of breast cancer, clear cell adenocarcinoma (CCA) of the vagina and cervix, and reproductive anomalies. This article will summarize the potential long-term health implications of DES, the role of nurse practitioners in identifying exposed individuals, and proper clinical management.

Six cases of women with diethylstilbestrol in utero demonstrating long-term manifestations and current evaluation guidelines.

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Diethylstilbestrol (DES), a nonsteroidal estrogen, was widely used in the United States from 1940 through 1971 to prevent pregnancy loss. In the late 1960s, an association was made with an increased incidence of clear cell adenocarcinoma in young women exposed in utero to DES. Additional study of these women over the next 35 years has shown an increased risk of other health problems including intraepithelial neoplasia, ectopic pregnancy, first trimester spontaneous abortion and second trimester pregnancy loss. The National Institutes of Health continues to fund studies to follow cohorts of DES-exposed mothers, daughters, sons and third generation children. The Centers for Disease Control have conducted a large DES Education Project and have established guidelines for management. The following six cases studies illustrate common problems seen in DES exposed daughters and management of problems encountered.

Diethylstilbestrol (DES): also harms the third generation.

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Diethylstilbestrol(DES) is a synthet- ic nonsteroidal oestrogen and endo- crine disruptor that was used in the 1950s-1970s to prevent spontaneous abortion, despite its lack of proven efficacy. Millions of women worldwide took DES during pregnancy. In France, between 1951 and 1981, about 160 000 children were exposed to DES during the first trimester of their intrauterine life, and in some cases almost throughout the entire pregnancy. They are referred to as "DES daughters" and "DES sons". In 2010, in France, about 25 000 DES daughters were aged 33 to 40 years: pregnancies among these women are foreseeable until about 2020. In utero exposure to DES can have harmful effects. In particular, DES daughters have an increased risk of cancer and structural abnormalities of the uterus that can adversely affect their pregnancies. What are the consequences of tak- ing DES during pregnancy for the third generation, i.e. the children of DES children? To answer this question, we reviewed the available data in mid- 2016 using the standard Prescrire methodology. According to a retrospective study conducted in France by Réseau DES France, published in 2016, which included 4409 DES grandchildren (2228 girls and 2181 boys) and about 6000 controls, about one-quarter of DES grandchildren are born prematurely. Preterm delivery exposes neonates to serious neonatal complications, including neurosensory disorders, disabilities and increased neonatal mor- tality. The more premature the baby, the greater the risk of complications. In the Réseau DES France study, cerebral palsy was more frequent in the DES grandchildren group: 59/10 000, versus 6/10 000 in the control group. A study conducted in the United States in about 4500 DES daughters found that preterm delivery occurred at a frequency of about 26%, much higher than that reported in controls. Neonatal mortality was 8 times higher among DES grandchildren, and the risk of stillbirth was twice as high. Other smaller studies have also shown an increased risk of preterm birth. A cohort study conducted in about 5000 DES grandchildren found that the risk of malformations of any type was higher than in the unexposed control group. Epidemiological studies, conduct- ed in several countries, found an increased frequency of hypospadias in DES grandsons. The relative risk was about 5 in the largest study. Other, less robust studies found no statistically significant difference. Several studies in several countries have shown a twofold increase in the risk of oesophageal atresia or tracheo-oesophageal fistula in DES grandchildren. The data on congenital heart defects or musculoskeletal malformations are limited and uninformative. Epidemiological studies have not identified a significant increase in the risk of gynaecological anomalies or cancers in DES granddaughters. Limited data are available on the risk of malformations in the children of DES sons. The data obtained in rodents exposed to DES (and other endocrine disruptors) make it entirely plausible that in utero exposure to DES, in humans too, provokes epigenetic effects that are passed on to future generations not directly exposed to DES. In practice, these data should be discussed with DES daughters, their partners and healthcare teams so that appropriate monitoring, clinical man- agement and follow-up can be arranged for both mother and baby. The harms of taking DES during pregnancy last for decades and affect future generations.

Long-term cancer risk in women given diethylstilbestrol (DES) during pregnancy.

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From 1940 through the 1960s, diethylstilbestrol (DES), a synthetic oestrogen, was given to pregnant women to prevent pregnancy complications and losses. Subsequent studies showed increased risks of reproductive tract abnormalities, particularly vaginal adenocarcinoma, in exposed daughters. An increased risk of breast cancer in the DES-exposed mothers was also found in some studies. In this report, we present further follow-up and a combined analysis of two cohorts of women who were exposed to DES during pregnancy. The purpose of our study was to evaluate maternal DES exposure in relation to risk of cancer, particularly tumours with a hormonal aetiology. DES exposure status was determined by a review of medical records of the Mothers Study cohort or clinical trial records of the Dieckmann Study. Poisson regression analyses were used to estimate relative risks (RR) and 95% confidence intervals (CI) for the relationship between DES and cancer occurrence. The study results demonstrated a modest association between DES exposure and breast cancer risk, RR = 1.27 (95% CI = 1.07-1.52). The increased risk was not exacerbated by a family history of breast cancer, or by use of oral contraceptives or hormone replacement therapy. We found no evidence that DES was associated with risk of ovarian, endometrial or other cancer.